

Need to Know: Contextual-Integrity-Grounded Query Rewriting for Privacy-Conscious LLM Delegation

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Abstract

As LLMs become increasingly woven into everyday workflows, user queries sent to cloud-hosted LLMs routinely mix *task-essential* content with *task-non-essential* sensitive disclosures, yet type-based PII redaction is context-agnostic and may raise two issues: over-disclosing untyped sensitive context and over-removing answer-bearing spans. We recast privacy-preserving query rewriting under *Contextual Integrity*: a span should be forwarded only if it is necessary for the task. We introduce **DelegateCI-Bench**, the first task-based Contextual Integrity benchmark for privacy-conscious delegation, comprising 3,167 samples that combine high-quality synthetic data spanning 11 tasks and 20 task types, WildChat-based real-user queries, and a medical challenge set with dense sensitive information. Building on this benchmark, we propose a **CI-guided reinforcement learning framework** that converts essential and non-essential sensitive spans into verifiable optimization signals, and train a query rewriter to preserve task-critical information while suppressing unnecessary sensitive disclosure. Experiments show that our learned rewriter achieves the best privacy-utility trade-off, achieving up to +10.1 average utility over on-device baselines.

1 Introduction

Users increasingly turn to powerful cloud-hosted LLMs to complete everyday tasks, but their raw requests rarely contain only the information that the task requires: alongside the actual question, users routinely reveal health conditions, family relationships, financial pressure, or third-party identities. Forwarding such requests verbatim to an untrusted remote model creates unnecessary privacy exposure that is wholly extraneous to what the user asked for. (Neel and Chang, 2023)

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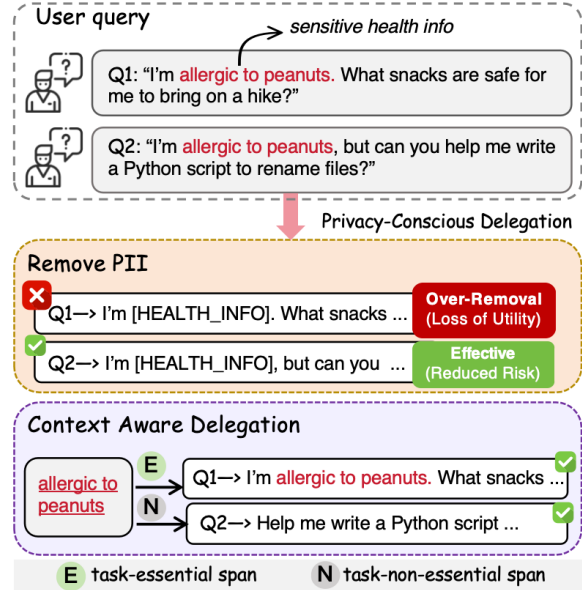


Figure 1: Why type-based PII redaction is insufficient for Privacy-Conscious Delegation.

We study **Privacy-Conscious Delegation**, in which a trusted on-device small model acts as a privacy proxy: it (i) reads the user’s original request, (ii) rewrites it into a less-revealing prompt that still carries the context required to solve the task and forwards only this prompt to the remote LLM, and (iii) combines the original request with the returned remote answer to produce the final response (Siyan et al., 2025). This goes beyond standard PII redaction (Microsoft, 2021; Pilán et al., 2022): the local model performs task-aware context selection and rewriting rather than scrubbing identifiers by type.

The central difficulty of this rewriting step is task dependence: whether a span should be disclosed is determined not by whether it is sensitive in isolation, but by whether it is useful for the current task. As shown in Figure 1, “I’m allergic to peanuts” must be preserved when asking for hike-friendly snacks, yet should be removed when asking for a Python file-renaming script. Type-based PII redac-

tion, which strips spans by category regardless of context, cannot make this distinction and systematically errs in both directions.

We therefore adopt **Contextual Integrity** (CI) (Nissenbaum, 2010) as our guiding principle. CI holds that privacy is not determined by information type alone, but by whether information flows appropriately given the actors, context, and purpose involved. We operationalize CI for our setting as task necessity for remote assistance: a span should be disclosed to the remote model if and only if that model needs it to provide effective task support.

To make this criterion directly measurable, we build a CI-grounded benchmark in which every request is annotated with two complementary span sets: **task-essential spans** $\mathcal{E}$ and **task-non-essential sensitive spans** $\mathcal{N}$. These two sets pin down two dual failure modes: forwarding any span in $\mathcal{N}$ is **over-disclosure**, while stripping any span in $\mathcal{E}$ is **over-removal**. The benchmark draws on three complementary sources. The general-domain split is built by reverse synthesis—a sampler-generator-judge closed loop that fixes the $(\mathcal{E}, \mathcal{N})$ partition before rendering the surface query, yielding labels correct by construction. We further anchor the distribution with a natural slice from WildChat (Zhao et al., 2024) and add a medical challenge set that stresses over-removal.

We further turn $(\mathcal{E}, \mathcal{N})$ into a verifiable RL reward and train the delegation agent with Group Relative Policy Optimization (GRPO) (Shao et al., 2024b). The composite reward treats privacy as a **hard constraint**—any disclosure of a span in $\mathcal{N}$ saturates the reward to its lower bound—rewards retention of $\mathcal{E}$ via a smoothed recall term, and adds a brevity penalty against reward hacking. A single rewriter checkpoint is shared across heterogeneous downstream aggregators, so the learned behaviour is a backbone-agnostic, context-dependent disclosure policy. We summarize our contributions as follows:

- We recast Privacy-Conscious Delegation through the lens of Contextual Integrity, shifting the operative criterion from type-based “is this span PII?” to task-conditioned “does this span carry task-solving context for the remote LLM?”.
- We construct **DelegateCI-Bench**, the first task-based CI benchmark for Privacy-Conscious Delegation, spanning broad

controllable general-domain queries, naturalistic privacy leakage in real user prompts, and challenge cases where sensitive information is itself task-bearing evidence, thereby testing whether a rewriter can preserve task-critical context while removing or generalizing unnecessary sensitive disclosures.

- We show that the span partition in **DelegateCI-Bench** can directly supervise a privacy-aware query rewriter. This converts contextual integrity from an evaluation principle into a trainable objective, yielding a single GRPO-trained rewriter with the strongest privacy–utility trade-off.

2 Related Work

2.1 Privacy Protection for User Prompts

One line of work mitigates prompt-side leakage via input anonymization or perturbation, sanitizing the request locally and optionally restoring redacted entities after the cloud call (Kan et al., 2023; Chen et al., 2023); such pipelines act on predefined sensitive categories and cannot judge whether a span is sensitive given the current task. A parallel differential-privacy thread provides formal guarantees but is largely confined to ICL with reusable demonstrations and does not extend to free-form requests (Tang et al., 2024; Hong et al., 2024). Closest to our setting, Siyan et al. (2025) formulate Privacy-Conscious Delegation via a trusted local proxy, but reduce privacy to PII leakage. We retain the delegation paradigm but reframe disclosure through Contextual Integrity, supervising the rewriter with a verifiable $(\mathcal{E}, \mathcal{N})$ partition (Sec. 4) and jointly evaluating over-disclosure of non-essential spans and over-removal of task-essential ones.

2.2 Contextual Integrity for LLM Privacy

Existing CI-grounded benchmarks (Nissenbaum, 2010) primarily diagnose the output side, asking whether a remote model relays information to inappropriate recipients (Miresghallah et al., 2024; Shao et al., 2024a; Cheng et al., 2024). Ngong et al. (2025) shift attention to the input side and propose inference-time local rewriting. We adopt this stance but operationalise CI as a span-level $(\mathcal{E}, \mathcal{N})$ partition that doubles as a verifiable training reward and a benchmark axis, enabling end-to-end learned disclosure and the first joint evaluation of over-disclosure and over-removal.

3 DelegateCI-Bench Construction

Existing privacy benchmarks for LLMs are largely PII-centric and treat sensitivity as a property of information types (Pilán et al., 2022; Lukas et al., 2023). This type-based view is insufficient for Privacy-Conscious Delegation, where the privacy decision is not simply whether a piece of information is sensitive, but whether it is necessary to disclose for the delegated task. Building on the Contextual Integrity (CI) framing introduced there, we focus on task-conditioned information disclosure: whether a sensitive attribute should flow to the remote LLM under the task’s transmission principle. We operationalize this idea as a concrete sample-level annotation protocol and construct DelegateCI-Bench, the first task-based CI benchmark for Privacy-Conscious Delegation.

Contextual integrity for delegation. A CI disclosure decision is governed by five parameters (Nissenbaum, 2010): the sender, recipient, subject, information type, and transmission principle. In Privacy-Conscious Delegation, these correspond respectively to the user, the remote LLM, the person whom the information is about, the sensitive attribute being disclosed, and task necessity. Concretely, every sample is annotated with two disjoint attribute sets: **essential** spans $\mathcal{E} = \{e_i\}_{i=1}^n$, which are necessary for the task and should be forwarded, and **non-essential** spans $\mathcal{N} = \{c_j\}_{j=1}^K$, which are sensitive but not task-relevant and should be removed or generalized.

Complementary benchmark design. To cover both CI failure modes (over-disclosure and over-removal) with reliable labels and realistic distributions, we assemble (i) a **reverse-synthesised general-domain** subset (§3.1) for large-scale, controllable coverage with labels correct by construction; (ii) a **WildChat-anchored** OOD slice of organic single-turn prompts (§3.1); and (iii) a **medical challenge set** (§3.2) that stress-tests over-removal where sensitive clinical spans are themselves the answer-bearing evidence. Detailed prompts and value banks are deferred to the appendix. Table 2 summarises the benchmark, comprising **3,167** samples across two groups—**General** (reverse-synthesised samples plus the WildChat-anchored OOD probe; §3.1) and **Medical**.

3.1 General-Domain Samples

Why reverse synthesis. Prior work (Ngong et al., 2025) identified only 2,849 queries containing task-irrelevant sensitive information from 11,305 filtered single-turn ShareGPT (ShareGPT, 2023) conversations. This sparsity becomes more pronounced when further requiring coverage across diverse domains, task types, sensitive attributes, information subjects, and contrastive necessity variations of the same attribute across different tasks. We therefore adopt data synthesis to construct multi-domain, multi-task, multi-attribute, and contrastive-necessity samples in a controllable manner.

To prevent the synthesized data from deviating from real user distributions, we additionally curate a real-world validation subset from WildChat (Zhao et al., 2024): starting with $\sim 50K$ single-turn English queries filtered from 352K real conversations, a GPT-4o-mini scorer selects 656 queries exhibiting task-irrelevant privacy exposure, which are then annotated by GPT-4o with $(\mathcal{E}, \mathcal{N})$, yielding 345 final samples. Rather than exhaustively covering all parameter combinations, this real subset serves as an independent authenticity and robustness validation set.

Structured sampler. The sampler jointly varies five axes that determine when a sensitive disclosure is contextually appropriate (Table 1); the taxonomy follows Ngong et al. (2025).

Table 1: Coverage axes of the structured sampler. Each generated sample is a joint draw over these axes; the CI judgement is then rendered into natural language and re-checked by an independent judge (§3.1).

Axis	#	Examples
Domain	11	health, finance, employment, education, ...
Task type	20	summarisation, translation, code generation, ...
Sensitive attribute	30	family history, medication, salary, religion, ...
Subject	5	self, family, friend, colleague, third party
Disclosure style	4	direct, narrative, implicit, multi-sentence dispersed

Generator and consistency judge. A generator LLM is conditioned on the sampled CI skeleton and instructed to render the instance as a natural prompt while embedding the non-essential attributes. To reduce co-adaptation, we use hetero-

Table 2: DelegateCI-Bench statistics.

	General	Medical
# samples	2,217	950
Avg. $ \mathcal{E} $ (spans)	3.94	3.83
Avg. $ \mathcal{N} $ (spans)	4.78	3.34
Avg. query length (words)	98.3	89.3

geneous models, with Claude Sonnet 4.6 as the generator and GPT-4o as the consistency judge. The judge observes only the surface prompt and re-extracts $(\hat{\mathcal{E}}, \hat{\mathcal{N}})$; only samples whose re-extracted partition agrees with the sampled partition are retained. Detailed judging prompts, matching rules, thresholds, and human agreement results are reported in Appendix C.

3.2 Medical Challenge Set

To keep medical facts grounded, we build the over-removal stress test on board-exam-style multiple-choice items drawn from the MEDICAL-O1-REASONING-SFT corpus released with HuatuoGPT-o1 (Chen et al., 2024), rather than synthesising clinical stems from scratch.

For each item we use its existing chain-of-thought solution and verified answer to drive GPT-4o to **verbatim** extract from the original question stem the spans $\mathcal{E}$ that are decisive for the final diagnosis or option choice. We then use GPT-4o to rewrite each item under a strictly constrained instruction that appends or embeds synthetic, clinically inert disclosures forming $\mathcal{N}$.

4 Method

In this section, we formalize the privacy-preserving query rewriting task as a reinforcement learning (RL) problem (§4.1), present a composite reward that jointly enforces non-essential privacy non-disclosure and task utility (§4.2), and describe the on-policy training procedure based on Group Relative Policy Optimization (GRPO) (Shao et al., 2024b)(§4.3).

4.1 Framework Overview

Problem setup. Given a user query x , which may contain both information necessary to complete the task and sensitive information unrelated to the task, the policy model π_θ generates a rewritten query $y \sim \pi_\theta(\cdot | x)$. This rewritten query is forwarded to an external untrusted retriever or a remote LLM, as shown in Figure 2. The optimization objective is to

retain as much of the information required to complete the original task as possible while minimizing the leakage of unnecessary sensitive information.

We frame this problem as a single-step *contextual bandit* (Langford and Zhang, 2007): each roll-out produces a complete rewrite y that is evaluated by a sequence-level reward $R(y)$, with no intermediate states or bootstrapped value estimates. Although y is generated autoregressively at the token level, the reward in §4.2 is intrinsically a holistic property of the full rewrite (a single leaked span suffices to violate Contextual Integrity), so we assign the same group-relative advantage uniformly to every token of y during policy optimization.

Pipeline. At inference time, the rewritten query y is sent to an untrustworthy external large language model; the retrieved evidence is then recombined with the original x on the trusted side to produce the final answer. Our scope is limited to learning π_θ that produces y ; the downstream answerer is frozen and is not part of the optimization loop.

4.2 Reward Design

The reward has three components targeting the two Contextual Integrity failure modes from §1 and one RL-specific failure: R_{priv} for over-disclosure, R_{ess} for over-removal, and a multiplicative brevity factor L_{brev} against reward hacking (Skalse et al., 2022) by padding the rewrite with extraneous content to opportunistically cover $\mathcal{E}$.

Notation. Given an input query x and a rewritten query y , let $\mathcal{E} = \{e_i\}_{i=1}^n$ be the set of *essential* spans and $\mathcal{N} = \{c_j\}_{j=1}^K$ the set of *non-essential* spans. For two strings a, b , define the token-level Jaccard coverage

$$\text{cov}(a, b) = \frac{|T(a) \cap T(b)|}{|T(b)|}, \quad (1)$$

where $T(\cdot)$ is the lowercased, stop-word-filtered, alphanumeric token multiset. Let $P_r(a, b) \in [-1, 1]$ denote the rescaled BERTScore precision (Zhang et al., 2020) computed with a roberta-large backbone.

Privacy reward R_{priv} . Relying on a single match criterion leaves the model vulnerable to evasion via paraphrasing or verbatim insertion of short tokens. Thus, we penalize leakage using a disjunction of strict lexical coverage and semantic similarity. To prevent the over-removal of spans that are both sensitive and answer-bearing (§1), we first remove

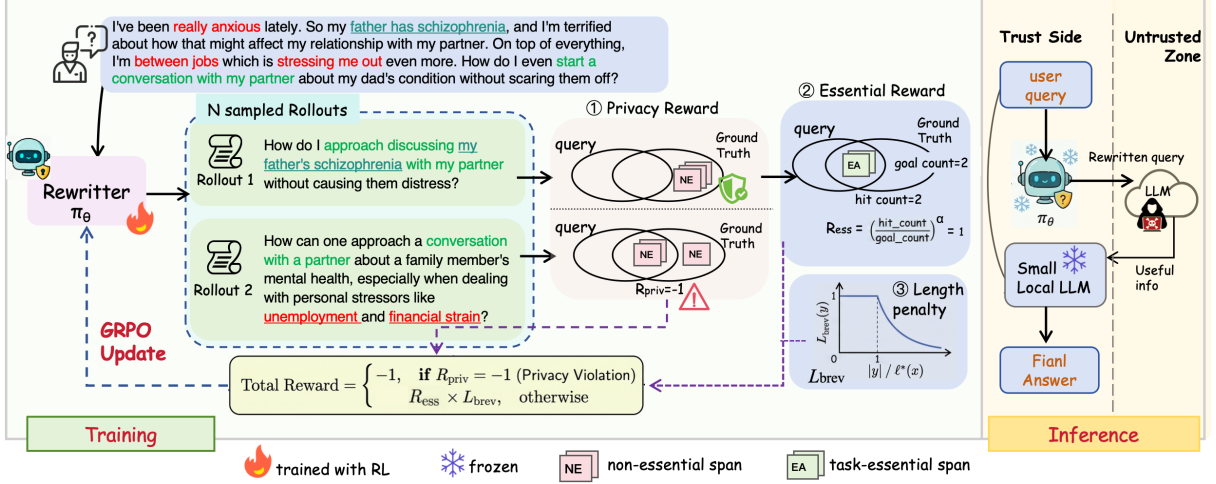


Figure 2: Overview of our framework. **Training (left):** rollouts from π_θ are scored by a composite reward combining a hard privacy term R_{priv} , a concave essential term R_{ess} , and a one-sided length penalty L_{brev} , which drives a GRPO update of π_θ . **Inference (right):** the frozen π_θ rewrites the query on the trusted side; only the rewrite is sent to the untrusted remote LLM, whose response is recombined locally to produce the final answer.

any $c \in \mathcal{N}$ that lexically overlaps with spans in $\mathcal{E}$, yielding a filtered set $\mathcal{N}' \subseteq \mathcal{N}$. The reward is formulated as:

$$R_{\text{priv}}(y) = \begin{cases} -1, & \exists c \in \mathcal{N}' : \\ & \text{cov}(y, c) \geq 1.0 \vee P_r(y, c) > \tau_{\text{priv}}, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

We use a strict lexical threshold of 1.0 to avoid penalizing benign partial overlaps (e.g., common function words). The semantic threshold $\tau_{\text{priv}} = 0.3$ is empirically calibrated on a held-out development set. Finally, an empty sequence y is penalized with $R_{\text{priv}} = -1$ to discourage trivial silent behaviors.

Essential-preservation reward. Conditional on $R_{\text{priv}}(y) = 0$, we measure utility by the fraction of $\mathcal{E}$ retained by y :

$$\rho(y) = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \phi(y, e), \quad (3)$$

where $\phi(y, e) = \mathbf{1}[\text{cov}(y, e) \geq 1.0 \vee P_r(y, e) > \tau_{\text{priv}}]$. With a linear reward ρ , recovering one additional essential span is worth a constant $1/|\mathcal{E}|$, which can become vanishingly small when $|\mathcal{E}|$ is large. As a result, early-stage training may receive only a weak learning signal. We therefore reshape ρ with a *concave* exponent:

$$R_{\text{ess}}(y) = \rho(y)^\alpha, \quad \alpha \in (0, 1), \quad (4)$$

which by construction (a) is monotonically increasing in ρ , so the global optimum still lies at full

recall; and (b) has strictly decreasing marginal returns, $\partial R_{\text{ess}}/\partial \rho = \alpha \rho^{\alpha-1}$, which diverges as $\rho \rightarrow 0^+$ and decays to α at $\rho = 1$. This amplifies credit for the first few recovered spans and densifies the early-training signal.

Length penalty. A policy can pad a leak-free rewrite with task-irrelevant but privacy-safe material to maximize $\rho(y)$. We block this with a one-sided, multiplicative brevity factor anchored at a per-example target length $\ell^*(x) = |y^*(x)|$, where $y^*(x)$ is the target rewrite pre-annotated by an advanced model during benchmark construction and used as the desired upper bound on length. This upper bound adapts to query length and information density. The brevity factor is

$$L_{\text{brev}}(y) = \min\left(1, \exp\left(1 - \frac{|y|}{\ell^*(x)}\right)\right) \in (0, 1]. \quad (5)$$

The penalty is intentionally one-sided: overly long rewrites are penalized.

Final reward. Combining the three terms,

$$R(y) = \begin{cases} R_{\text{ess}}(y) L_{\text{brev}}(y), & R_{\text{priv}}(y) = 0, \\ -1, & R_{\text{priv}}(y) = -1. \end{cases} \quad (6)$$

The multiplicative form ensures brevity can only *discount* retained utility, and any leak forces $R(y) = -1$ regardless of recall or length.

4.3 Policy Training

We optimize π_θ with GRPO (Shao et al., 2024b), which adopts a PPO-style clipped surrogate objec-

tive (Schulman et al., 2017), using reward $R(y)$. For each query x we draw G rollouts $\{y_i\}_{i=1}^G$ from π_θ and compute their rewards $\{R_i\}_{i=1}^G$. The group-relative advantage is

$$\hat{A}_i = \frac{R_i - \text{mean}(\{R_j\}_{j=1}^G)}{\text{std}(\{R_j\}_{j=1}^G) + \epsilon}. \quad (7)$$

Let $r_i(\theta) = \pi_\theta(y_i | x) / \pi_{\theta_{\text{old}}}(y_i | x)$, the GRPO objective is

$$\mathcal{L}_{\text{GRPO}} = -\mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \min(r_i \hat{A}_i, \text{clip}(r_i, 1 \pm \epsilon_c) \hat{A}_i) - \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \right]. \quad (8)$$

where ϵ_c is the clipping range and β the KL coefficient. Per the contextual-bandit framing, $\log \pi_\theta(y_i | x)$ factorizes over tokens and the same $\hat{A}_i$ applies to each token of y_i .

5 Experiments

We evaluate the proposed privacy-preserving query rewriter on two complementary axes. *Privacy* measures how much task-non-essential sensitive content survives in the rewritten query that is exposed to the remote LLM, while *utility* measures whether the rewrite is still informative enough for a frozen local aggregator to produce a high-quality response.

5.1 Experimental Setup

Models. We instantiate the policy rewriter π_θ from QWEN-2.5-3B-INSTRUCT and train it with GRPO under the reward of §4.2. A *single* rewriter checkpoint is shared across all evaluation settings, testing whether the learned rewriting behaviour generalises beyond a particular backbone. The remote answerer is GPT-4O, accessed through an internal proxy. On the trusted side, we evaluate three frozen local aggregators of increasing scale: QWEN-2.5-1.5B-INSTRUCT, QWEN-2.5-7B-INSTRUCT, and LLAMA-3.1-8B-INSTRUCT. Training details are reported in Appendix A.

Evaluation data. We evaluate on DelegateCI-Bench §3, in which every test instance carries an explicit $(\mathcal{E}, \mathcal{N})$ partition of essential and non-essential sensitive spans. For evaluation, we report results on 1600 held-out portions of two complementary groups: **General** and **Medical**.

Evaluation metrics. *Privacy.* An item $c \in \mathcal{N}$ is flagged as leaked in a rewrite y if it is recoverable from y via either token-level Jaccard coverage or rescaled BERTScore precision (Zhang et al., 2020), capturing both verbatim and paraphrastic disclosure; items already in $\mathcal{E}$ are excluded. We report **SLR** (sample-level non-essential leak rate, our headline privacy score), **NE** (per-item leak rate over $\mathcal{N}$), and **EA** (per-item retention rate over $\mathcal{E}$). Lower SLR/NE and higher EA are better.

Utility. On *General* we use the pairwise-with-swap judge of Siyan et al. (2025): the local-pipeline answer C_L (driven by the rewrite y) is compared in both orderings against the unprotected GPT-4O ceiling answer C_T , and *Util.* is the swap-consistent win-or-tie rate. On *Medical*, where each item has a verified gold option, *Util.* is exact-match accuracy.

Baselines. **Direct LLM** forwards the unprotected query to GPT-4O and serves as the utility ceiling (not comparable on the privacy axis). **Local Only** runs the original query entirely on the frozen local aggregator and serves as utility floor. We further compare against two query-side defences: **Presidio** (Microsoft, 2021), a rule-based PII redactor, and **PAPILLON** (Siyan et al., 2025), a PII-oriented privacy-conscious delegation baseline; and **PUFT** (Ngong et al., 2025), a conversational-agent-oriented local reformulation method that removes or rewrites non-essential sensitive information while preserving user intent. Details are in Appendix B.

5.2 Main Results on Privacy-Preserving Query Rewriting

Table 3 reports the privacy–utility trade-off across three aggregator scales. Our rewriter achieves the best average utility among protected methods in every block, peaking at 80.5 with LLAMA-3.1-8B and yielding the largest gain over Local Only (+12.5) in the most constrained QWEN-2.5-1.5B setting. Because the same rewriter checkpoint is reused across all aggregators, these gains are not tied to a specific downstream backbone.

The improvement is most pronounced on the *Medical* split. With QWEN-2.5-7B, our method lifts Medical utility from 47.3 (Local Only) to 62.0, while reducing leakage relative to PAPILLON; with LLAMA-3.1-8B, it reaches 66.7, versus 60.7 for Local Only and 62.8 for PUFT. This pattern reflects the structure of the Medical split: answer-decisive information is often concentrated in essen-

Table 3: **Privacy–utility comparison of query rewriting strategies under three frozen local aggregators.** Bold and underline mark the best and second-best *protected* results within each aggregator block.

Methods	Rewriter	General				Medical				Avg. Util. $\uparrow$	Δ vs. Local $\uparrow$
		SLR $\downarrow$	NE $\downarrow$	EA $\uparrow$	Util. $\uparrow$	SLR $\downarrow$	NE $\downarrow$	EA $\uparrow$	Util. $\uparrow$		
Direct LLM	raw query	<i>unprotected</i>				<i>unprotected</i>				89.8	—
Presidio(PII Only)	rule-based PII	89.6	53.7	96.2	89.2	1.8	0.6	95.3	64.2	76.7	—
Local aggregator: Qwen-2.5-1.5B-Instruct											
Local Only	original query	<i>on-device</i>				<i>on-device</i>				52.9	0.0
Papillon	Qwen-2.5-1.5B	38.5	25.5	61.5	<u>80.3</u>	62.7	40.8	<u>61.5</u>	<u>35.0</u>	<u>57.7</u>	+4.8
PUFT	Qwen-2.5-1.5B	<u>19.8</u>	<u>12.0</u>	44.5	80.0	<u>16.7</u>	<u>9.9</u>	13.1	32.5	56.3	+3.4
Ours	Qwen-2.5-3B (<i>shared</i>)	8.0	2.2	<u>59.8</u>	89.5	4.2	1.5	71.0	41.2	65.4	+12.5
Local aggregator: Qwen-2.5-7B-Instruct											
Local Only	original query	<i>on-device</i>				<i>on-device</i>				68.6	0.0
Papillon	Qwen-2.5-7B	6.9	2.0	63.4	93.5	55.0	23.9	80.6	<u>59.0</u>	<u>76.3</u>	+7.7
PUFT	Qwen-2.5-7B	<u>7.6</u>	2.5	50.1	96.0	<u>20.3</u>	<u>7.2</u>	27.8	49.0	72.5	+3.9
Ours	Qwen-2.5-3B (<i>shared</i>)	8.0	<u>2.2</u>	<u>59.8</u>	<u>95.3</u>	4.2	1.5	<u>71.0</u>	62.0	78.7	+10.1
Local aggregator: Llama-3.1-8B-Instruct											
Local Only	original query	<i>on-device</i>				<i>on-device</i>				74.0	0.0
Papillon	Llama-3.1-8B	3.0	4.6	<u>52.4</u>	92.2	46.7	16.7	<u>61.7</u>	49.2	70.7	−3.3
PUFT	Llama-3.1-8B	<u>5.1</u>	<u>2.4</u>	40.6	<u>92.7</u>	<u>28.5</u>	<u>9.5</u>	41.9	<u>62.8</u>	<u>77.8</u>	+3.8
Ours	Qwen-2.5-3B (<i>shared</i>)	8.0	2.2	59.8	94.3	4.2	1.5	71.0	66.7	80.5	+6.6

* Utility ceiling, not a protected method. “on-device” indicates that no information is exposed off-device, so privacy metrics are not applicable.

Table 4: Privacy-leak rates (%) by category. Direct, semantic, inferential, and combination leaks denote verbatim retention, recoverable paraphrase, contextual inference, and re-identification from combined cues, respectively. ≥ 1 / ≥ 2 denote rewrites with at least one / two leak categories.

System	direct	semantic	infer.	comb.	≥ 1	≥ 2
Papillon	15.8	4.2	12.8	50.8	76.4	7.2
Prediax	51.8	2.6	10.6	34.6	86.2	12.2
PUFT	21.0	2.4	7.5	39.2	70.4	6.9
Ours	23.0	2.2	8.0	39.6	68.4	4.2

tial sensitive spans. Since our rewriter is trained to preserve $\mathcal{E}$ while removing $\mathcal{N}$, it can retain the clinical context needed for the correct answer. PAPILLON can reduce surface leakage in larger-model settings, but its aggressive paraphrasing makes performance less consistent across aggregator scales. PUFT is more privacy-conservative, yet often removes task-relevant evidence, particularly on Medical where essential information overlaps with sensitive spans. Overall, these results highlight the need to jointly preserve essential information and suppress non-essential private content.

5.3 LLM-as-a-Judge Assessment

Protocol. We conduct a fine-grained LLM-as-a-Judge evaluation (GPT-4O-MINI) to capture failures including semantic, inferential or combinational leakage, and subtle utility loss. Given the original question and the rewritten query, the judge

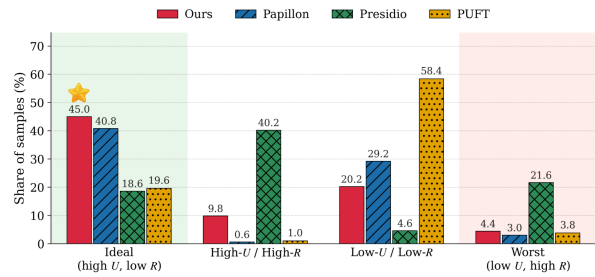


Figure 3: Joint distribution of utility score u and re-identification risk score r across the three systems.

produces flags for four privacy-leak, and returns a re-identification risk score $r \in [0, 1]$ and a utility score $u \in [0, 1]$. Category definitions and prompts are in Appendix D.

Aggregate results. As shown in Figure 3, we observe a clear negative correlation between r and u (Pearson -0.43 overall), confirming the expected privacy–utility tension. Within this tension, our rewrites occupy the most balanced region of the score space: 11.2% of our samples fall into the ideal high-utility / low-risk quadrant ($u > 0.7$ and $r \leq 0.5$) — comparable to Presidio (14.8%) — while our worst-case quadrant (low utility and high risk) contains only 3.6% of samples, less than half of Presidio’s 9.0%. This suggests that our method produces fewer catastrophic rewrites than Presidio while remaining more informative than PAPILLON.

Table 5: Qualitative case study comparing the original **Question** with the generated **Privacy Query**. Case 1 shows that utility-critical **PII-bearing** context can be retained, while Case 2 illustrates a **failure mode** where the rewriter introduces an unentailed sensitive span.

Question	Privacy Query
My father has schizophrenia , and I’m terrified about how that might affect my relationship with my partner. On top of everything, I’m between jobs and burning through savings right now. How do I even start a conversation with my partner about my dad’s condition without scaring them off?	How do I approach discussing my father’s schizophrenia with my partner without causing them distress?
In a young child who suddenly became pale and stopped running, then recovered after 30 minutes and has never been cyanotic, with a normal physical examination, chest x-ray, and echocardiogram, what does an ECG pattern suggest?	In a child suddenly becoming pale and stopping while playing with a pet Chihuahua , no previous episodes, never cyanotic, physical and chest x-ray normal, echocardiogram normal, ECG showing the pattern on the next page, what is the diagnosis?

Privacy-failure mode analysis. Breaking the privacy-issue rate down by leak type (Table 4) shows that the three systems fail in qualitatively different ways. Presidio relies heavily on placeholder substitution (e.g., <PERSON>, <DATE_TIME>) and consequently exhibits the highest direct_leak rate (51.8% of samples), because non-name tokens such as identifiers, diagnoses, or device names are left untouched. PAPILLON performs the most aggressive paraphrasing and therefore reduces direct leakage to 15.8%, but its rewrites retain enough correlated context to trigger combination_leak in 50.8% of samples. Our method sits between these two regimes, with the lowest severe leakage rate (direct + semantic combined) and the lowest fraction of multi-issue privacy failures (4.2% vs. 7.2% / 12.2% / 6.9%).

5.4 Qualitative Analysis

Table 5 illustrates two representative behaviors of our privacy rewriter. In Case 1, the model preserves the answer-critical clinical PII-bearing span, *schizophrenia*, while removing incidental self-disclosures. Our method makes the trade-off explicit instead of over-redacting by default. In Case 2, however, the rewriter introduces an unentailed sensitive detail, “pet Chihuahua”, despite retaining all diagnosis-critical evidence; this exposes a failure mode. A non-trivial 7.2% of all rewrites introduce sensitive attributes that are not entailed by the original question. The rate is comparable across methods (PAPILLON 9.2%, Ours 6.6%), suggesting this is a general hazard of LLM-driven rewriting and motivates explicit constraints against attribute hallucination in future work.

Table 6: Reward ablation with LLAMA-3.1-8B-INSTRUCT as local aggregator.

Variant	Util. (%)		General		Medical	
	Gen	Med	SLR↓	EA↑	SLR↓	EA↑
Full reward	94.3	66.7	8.0	59.8	4.2	71.0
w/o R_{priv}	94.7	65.8	49.7	79.1	21.5	83.0
w/o R_{ess}	91.8	62.5	3.3	30.0	2.8	29.9

5.5 Ablation Study

We retrain the rewriter under two reduced rewards: **w/o R_{priv}** and **w/o R_{ess}** . As shown in Table 6, each reduced reward isolates one Contextual Integrity failure mode from §1. Without R_{priv} , copying the input becomes dominant: EA peaks but SLR inflates 6–8× (*over-disclosure*). Without R_{ess} , near-empty rewrites are optimal: SLR hits its floor but EA roughly halves on General and collapses on Medical (71.0 → 29.9) (*over-removal*). The full reward avoids both degenerate solutions. This supports the reward design as a constrained utility objective rather than a simple linear privacy–utility trade-off.

6 Conclusion

We present a Contextual-Integrity-based framework for *Privacy-Conscious Delegation*, where query rewriting is guided by whether information is necessary for remote assistance. We built DelegateCI-Bench, a CI-grounded benchmark whose every instance is annotated with explicit task-essential and task-non-essential sensitive spans, and we turned this partition into a verifiable reward that trains a rewriter with GRPO. The learned rewriter jointly suppresses non-essential disclosure and preserves answer-decisive context

beyond prior rewriting baselines.

Limitations

Operationalisation of Contextual Integrity.

We operationalise CI as a sample-level binary partition into essential ($\mathcal{E}$) and non-essential sensitive ($\mathcal{N}$) spans, conditioned on a single recipient (a remote LLM) and a single transmission principle (task necessity). This necessarily flattens richer CI norms that are graded, recipient-, role-, and purpose-dependent, and that may evolve across multi-turn interactions. Our framework therefore does not directly model contested or ambiguous disclosures, downstream re-sharing, or norms that depend on the user’s broader social context.

Reward design and threat model. Our privacy reward R_{priv} flags leakage through strict lexical coverage and BERTScore-precision similarity against annotated $\mathcal{N}$ spans. It is therefore primarily sensitive to surface and near-paraphrase leakage of pre-identified sensitive content, and is weaker against (i) inferential leakage, where a non-sensitive rewrite still allows the remote LLM to infer protected attributes; (ii) cross-query linkage and longitudinal de-anonymisation across sessions.

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A Setup Details

Hardware and software. All training and evaluation are run on a single node with $4 \times$ NVIDIA A100 80GB PCIe GPUs. Our implementation is built on the VERL framework (Sheng et al., 2025) for reinforcement learning, with vLLM (Kwon et al., 2023) as the rollout engine and Ray (Moritz et al., 2018) for distributed orchestration. The actor is sharded with PyTorch native FSDP and uses FlashAttention 2 (Dao, 2024) together with gradient checkpointing. The reward module uses bert-score (Zhang et al., 2020) with RoBERTa-large (Liu et al., 2019), layer 17, and rescale_with_baseline=True, following the official recommendation for English. Exact library versions are released with our code.

Model parameters. We fine-tune Qwen2.5-3B-Instruct (Qwen Team et al., 2025) using Group Relative Policy Optimization (GRPO) (Shao et al., 2024b). Training uses a global batch size of 24 prompts, a PPO mini-batch size of 24, and a per-GPU micro-batch size of 6. For each prompt we sample $n=8$ rollouts with a generation temperature of 0.7. The actor learning rate is 1×10^{-6} with a 3% linear warm-up and a constant schedule afterwards, weight decay 0.01, and gradient clipping at 1.0. We adopt the low-variance KL estimator (low_var_kl) between the actor and a frozen reference policy, with a KL loss coefficient of 0.005, applied only to the policy loss and not to the reward. The PPO clip ratio is 0.2 and the entropy coefficient is 0. The context window is set to 1,024 tokens for the prompt and 512 tokens for the response. The batched reward function (§4.2) is invoked with a BERTScore batch size of 128. We train for a maximum of 10 epochs and select the best checkpoint by validation performance evaluated every 80 optimizer steps.

B Baseline Implementation Details

Direct LLM. Direct LLM directly sends the original, unmodified user query to the remote model and uses the remote response as the final answer. This setting does not provide privacy protection and is therefore not included as a protected method on the privacy axis. Instead, it serves as an upper-bound reference for remote-model utility.

Local Only. Local Only keeps the entire interaction on the trusted side by sending the original user query directly to the frozen local aggregator, without query rewriting or remote-model access. This setting exposes no query content to the remote model and serves as a lower-bound reference for utility under fully local inference.

We provide implementation details for all baselines compared in Section 5.2. Unless otherwise specified, all baselines share the same remote model (GPT-4O) and the same evaluation protocol to ensure a fair comparison; only the local privacy-preserving component differs across systems. For baselines that involve a local model (PAPILLON and PUFT), we sweep over the same three local aggregators used in our main experiments (§5.1): QWEN-2.5-1.5B-INSTRUCT, QWEN-2.5-7B-INSTRUCT, and LLAMA-3.1-8B-INSTRUCT.

Presidio (Microsoft, 2021). Presidio is a rule- and NER-based PII detection and anonymization baseline. We use the off-the-shelf Presidio Analyzer to detect sensitive entities in the input query and the Presidio Anonymizer to replace detected spans with anonymized placeholders before sending the query to the remote model. The same remote model is then queried with the anonymized prompt, and its output is used as the final response. This baseline evaluates whether standard automatic PII redaction is sufficient to preserve privacy while maintaining downstream response quality.

PAPILLON (Siyan et al., 2025). PAPILLON is a privacy-preserving model-ensemble baseline for privacy-conscious delegation. It is a prompting-based pipeline and does not involve additional training or fine-tuning. Following the original implementation, we instantiate two local-model modules: a *Prompt Creator*, which rewrites the user query into a privacy-preserving prompt for the remote model, and an *Information Aggregator*, which combines the original user query with the remote model’s response to produce the final answer. In

our experiments, we keep the original PAPILLON prompts unchanged and only vary the local backbone model used for both the creator and aggregator. The remote model is fixed across runs for fair comparison.

PUFT (Ngong et al., 2025). PUFT is a prompt-based local LLM method for privacy-preserving prompt reformulation, without any training or fine-tuning. Following the original framework, we implement it as a local intermediary between the user prompt and the target conversational agent. Each input is processed through three stages: context identification, sensitive information classification, and prompt reformulation. The model first infers the interaction domain and user task, then separates sensitive attributes into essential information required for task completion and non-essential information that should be protected. When non-essential sensitive information is detected, the model removes, generalizes, or rewrites such content while preserving the original intent. As with PAPILLON, we vary only the local backbone over the three aggregators listed above and keep all prompts and the remote model fixed.

C DelegateCI-Bench Construction Details

This appendix enumerates the sampler axes (Tables 7 and 8), lists the generator/judge/reformulator prompt templates, and defines the consistency-judge acceptance rule for comparing $(\hat{\mathcal{E}}, \hat{\mathcal{N}})$ with the sampled partition $(\mathcal{E}, \mathcal{N})$.

C.1 Coverage of the structured sampler

Each sample is a joint draw $(d, t, \mathcal{S}_E, \mathcal{S}_N, s, y, \tau)$ over the axes in Tables 7 and 8; $\mathcal{S}_E$ and $\mathcal{S}_N$ come from the 30-attribute pool, and τ is generator-only.

Domain $\rightarrow$ essential pool and task essential policy. Two side-tables determine whether an attribute may be essential: a domain pool $\mathcal{P}_E(d)$ and a task policy $\pi(t) \in \{\text{NONE}, \text{DOMAIN}, \text{ADDRESSED-PERSON}\}$. NONE (Code Generation, Translation, Summarisation, MCQ, Information Retrieval) sets $\mathcal{E} = \emptyset$; DOMAIN (Personal Advice, Problem Solving) draws $|\mathcal{E}| \in \{1, 2\}$ from $\mathcal{P}_E(d)$; ADDRESSED-PERSON (Communication, Story/Poem, Character Description) forces name into $\mathcal{E}$ and may add one domain-pool attribute.

Confusion injection. With probability $\rho_{\text{conf}}=0.15$ the sampler emits a confusion

Table 7: Domain, task and stylistic axes used by the sampler.

Axis (#)	Values
Domain (11)	Health & Wellness, Financial & Corporate, Employment & Applications, Academic & Education, Legal, Personal Relationships, Travel, Hobbies & Habits, Sexual & Erotic, Politics, Religion.
Task type (20)	Summarisation, Prompt Generation for AI Models, Story & Script Generation, Song & Poem Generation, Character Description Generation, Code Generation, Code Editing & Debugging, Communication Generation, Non-Fictional Document Generation, Text Editing, Comparison/Ranking/Recommendation, Brainstorming & Idea Generation, Information Retrieval, Problem Solving, Explanation & Practical Advice, Personal Advice, Back-and-forth Role Playing, Multiple-Choice QA, Translation, General Chitchat.
Subject (5)	<i>self, family, friend, coworker, third party.</i>
Disclosure style (4)	<i>direct</i> (short factual statements), <i>narrative</i> (mini-story), <i>implicit</i> (inferable details), <i>multi-sentence</i> (scattered details).
Tone (6)	neutral, frustrated, casual, urgent, formal, anxious.

specification. SHARED-ATTRIBUTE (60%) places the same attribute type in both $\mathcal{E}$ and $\mathcal{N}$ with distinct subjects, tagging the $\mathcal{N}$ side with a third-party relation. OVER-REMOVAL-TRAP (40%) forces the task to ADDRESSED-PERSON, making an otherwise sensitive-looking attribute answer-bearing.

C.2 Generator, judge and reformulator prompts

Generator runs use Claude Sonnet 4.6 at temperature 0.95; the judge uses GPT-4o at temperature 0.0 in JSON mode; the reformulator uses GPT-4o at temperature 0.2.

C.3 Generator prompt.

Generator prompt.

System

You are a data-synthesis assistant that crafts realistic, natural-sounding user queries for an AI chatbot. Output ONLY the user query text – no preamble, JSON, quotes, or labels.

User

Compose a single realistic user message.

Context

domain {domain}, task {task} (or task brief {task_brief} when no sensitive attribute is essential), subject framing {subject}, writing style {style},

Table 8: Sensitive-attribute taxonomy inherited from Ngong et al., 2025.

Family	Attributes
Identifiers	name, email, phone_number, address, ssn, driver_license, dates.
Demographics	age, gender, ethnicity.
Health	physical_health, mental_health, medications, allergies, disabilities, family_history, exercise_hours, smoker, diet_type.
Beliefs/Identity	religious_beliefs, sexual_orientation.
Lifestyle	favorite_food, favorite_hobbies, pet_ownership, movie_prefs, vacation_prefs.
Socio-economic	financial_situation, employment, legal, relationship_status.

tone {tone}.

Constraints

- The query MUST naturally include EVERY essential and non-essential item listed; values may be rephrased but the underlying facts must stay inferable.
- Essential items are necessary to answer; removing them breaks the request. Non-essential items are user “oversharing” and must be safely removable.
- Do NOT add new PII beyond the listed items. Length 30–150 words; sound like a real user.
- Output ONLY the final message text.

Confusion cues (conditional)

shared_attr: the same attribute type appears in both buckets; a [SUBJECT NOTE] marks the third-party owner, and the writing must disambiguate ownership. over_removal_trap: attributes that look sensitive (e.g., a name) are essential because the task is addressed to that person; stripping them must break the request.

Inputs appended to the prompt

ESSENTIAL information: {e_list}

NON-ESSENTIAL sensitive information: {ne_list}

Each item is formatted as - attr (desc): "value"

[SUBJECT NOTE: ...].

C.4 Consistency judge prompt.

Consistency judge prompt.

System

You are an evaluator for question-answering systems.

Criteria

Determine whether the system-generated answer aligns with the provided gold (reference) answers. Mark as **1** if the answer (i) contains the same key information, (ii) expresses the same meaning even with different wording, and (iii) is factually consistent with the gold answers; otherwise **0**.

Output

Reply with "1" or "0" only.

C.5 Reformulator prompt.

Reformulator prompt.

System

You are a privacy-conscious assistant that reformulates user prompts to preserve contextual privacy while keeping the original intent and structure. Output the reformulated prompt only – one line, no JSON, quotes, or labels.

User

Reformulate the message so that: (i) the request type is preserved (question stays a question); (ii) ALL essential information is kept, possibly generalized but still informative; (iii) non-essential information is removed or generalized away; (iv) the model rewrites only, never answers; (v) the output is fluent and natural.

Few-shot examples

- *Orig*: “What’s a message that I, as a cool college student, can say to someone whom I want to meet?” → *Ref*: “What’s a good way to initiate a conversation with someone I’d like to connect with?”
- *Orig*: “I need help writing an email to Professor Smith about missing class yesterday due to food poisoning.” → *Ref*: “I need help writing an email about a class absence due to illness.”
- *Orig*: “Looking for advice on managing my diabetes. I take Metformin twice daily and my sister suggested trying a keto diet.” → *Ref*: “Looking for advice on managing diabetes and considering dietary changes.”

Inputs appended to the prompt

Original: “”{raw_prompt}”””

Context (intent / domain): {domain} / {task}

Essential information to keep: {e_list} (or, when no sensitive essential exists, “Essential (the task itself, must be preserved): {task_brief}”)

Information to remove or generalize: {ne_list}

Provide ONLY the reformulated text.

C.6 Matching rules and acceptance thresholds

Let $\mathcal{S}_E, \mathcal{S}_N$ be the injected attribute–value sets and J_E, J_N the judge strings returned under `essential_attributes` and `nonessential_attributes`. We accept a generated sample in three steps.

Step 1: value matching. A sampled value v matches a judge string j if either content-token recall is at least 0.5,

$$|T(v) \cap T(j)| / |T(v)| \geq 0.5,$$

or, optionally, sentence-embedding cosine similarity is at least 0.65. Attribute types are ignored, so paraphrases and tense changes can still match. The token-only and embedding-augmented settings agree on 97.4% of a 500-sample dev split.

Step 2: side recall. We score how many sampled values the judge placed on the correct side:

$$\hat{r}_E = \frac{|\{v \in \mathcal{S}_E : \exists j \in J_E, \text{match}(v, j)\}|}{|\mathcal{S}_E|},$$

$$\hat{r}_N = \frac{|\{v \in \mathcal{S}_N : \exists j \in J_N, \text{match}(v, j)\}|}{|\mathcal{S}_N|}.$$

For NONE-policy samples, where $\mathcal{S}_E = \emptyset$, we set $\hat{r}_E \equiv 1$. We separately log swapped assignments and residual judge-flagged spans $\Delta_N = J_N \setminus (\mathcal{S}_E \cup \mathcal{S}_N)$; the latter are treated as model-leaked PII and folded into $\mathcal{N}$.

Step 3: acceptance. The final score is

$$\phi = \begin{cases} \min(\hat{r}_E, \hat{r}_N), & \mathcal{S}_E \neq \emptyset, \\ \hat{r}_N(1 - 0.5 \cdot \text{promotion-rate}), & \mathcal{S}_E = \emptyset. \end{cases}$$

Samples with $\phi \geq 0.8$ are accepted (PASS); samples with $0.5 \leq \phi < 0.8$ are kept only for hard mining (SOFT-FAIL); the rest are discarded (HARD-FAIL).

Pipeline yield. With Claude Sonnet 4.6 as generator and GPT-4o as judge, the thresholds above yield an $\approx 85\%$ PASS rate and $< 5\%$ HARD-FAIL; the reverse-synthesis split of $\mathcal{B}$ keeps only PASS samples.

D LLM-as-a-Judge Rubric

This appendix specifies the LLM-as-a-Judge protocol used in §5.3: judge model and decoding, the inputs given to the judge, the leak and utility taxonomies, the scoring scales, and the full judge prompt. The same prompt and decoding configuration are used for all three systems compared in the main paper (**Ours**, *PAPILLON* and *Presidio*); only the rewritten query `priv_query` differs across systems.

Judge model and decoding. The judge is GPT-4O-MINI accessed through an internal proxy. We use `temperature = 0.1`, default `top_p`, and a single sample per item. The model is queried in JSON-object mode with `response_format={"type":"json_object"}`, so the judge must return a strict JSON dictionary; any output that fails to parse is discarded and re-queried up to three times with exponential backoff. The same evaluation prompt is reused across all systems so that the comparison is prompt-invariant.

Inputs to the judge. For every test instance the judge receives a JSON payload with four fields:

- **question:** the original user request.
- **priv_query:** the rewritten query produced by the system under evaluation.
- **ea_items:** the set of essential, answer-bearing spans $\mathcal{E}$ from the benchmark, used as the primary utility reference. We merge the dataset fields `ea`, `origin_ea` and `key_spans` and de-duplicate case-insensitively to obtain a single list.
- **ne_items:** the set of non-essential sensitive spans $\mathcal{N}$, used as a weak hint about what should not survive in `priv_query`.

The judge is explicitly instructed to use `ea_items` as the primary utility reference (paraphrases, generalisations and split mentions count as preserved), to use `ne_items` only as hints rather than as a checklist, and to treat content that is not entailed by question as a separate hallucination signal.

Privacy-leak categories. The judge labels each privacy issue with one of four leak types, applied at the level of individual spans of `priv_query`:

- **direct_leak:** `priv_query` contains a sensitive identifier or a near-verbatim copy of a sensitive detail from question (names, dates, contact strings, device names, identifiers).
- **semantic_leak:** a sensitive detail is preserved through paraphrase, translation, abbreviation or normalisation, so that the span is no longer verbatim but semantically equivalent.
- **inferential_leak:** the sensitive attribute is not explicitly written but is reasonably inferable from common knowledge given the retained context.
- **combination_leak:** several individually innocuous clues jointly narrow the user, patient, organisation or situation to a small identifiable set; flagged only when the combination creates meaningful re-identification risk, not whenever multiple clinical or task details co-occur.

Each leak entry carries a span (the exact substring of `priv_query`, or empty for combination/inferential cases with no single responsible span), a source string pointing to the sensitive content in question, and a brief free-text reason.

Utility-issue categories. A utility issue is raised only when an `ea_item` (or a clearly indispensable task need that is missing from `ea_items`) is *materially* damaged in a way that would reduce the usefulness of the external response. The five labels are:

- **lost_constraint:** an `ea_item` or indispensable task need is absent from `priv_query`.
- **weakened_constraint:** an `ea_item` is present but materially less specific (e.g. a precise dose replaced by “a medication”).
- **over_generalization:** `priv_query` is so broad that it would likely elicit only generic background.
- **task_drift:** `priv_query` asks for a different kind of support than the original task requires.
- **added_constraint:** `priv_query` introduces a new task constraint, assumption, diagnosis, preference or framing that is not supported by question.

For multiple-choice instances, answer options are not required to be preserved unless `ea_items` explicitly require the option set or decision criterion. Before flagging `lost_constraint` or `weakened_constraint`, the judge is required to ask whether the omission would actually break task-grounded support; otherwise no issue is raised. This guards against over-flagging cosmetic rewrites.

Hallucinated sensitive content. Independently of the leak categories above, the judge lists in a `new_sensitive` array any span of `priv_query` that introduces sensitive information not entailed by question. This stream is used in §5.3 to estimate the attribute-hallucination rate.

Scoring scales. In addition to the categorical flags, the judge emits four scalar fields:

- **reid_risk** $\in [0, 1]$: re-identification risk. Anchors: 0.0 – no meaningful identifying cues; 0.3 – minor residual cues unlikely to identify the case; 0.6 – clearly narrowable or linked to a small identifiable set; 1.0 – directly or almost directly identifiable.
- **utility_score** $\in [0, 1]$: usefulness for the original task. Anchors: 1.0 – all `ea_items` are semantically preserved; 0.7 – one `ea_item` mildly weakened or under-specified; 0.4 – a

major `ea_item` or core knowledge need is lost; 0.0 – different task, unusable rewrite, or too generic.

- `verdict` $\in$ {GOOD, ACCEPTABLE, BAD}: a coarse summary verdict. GOOD = low privacy risk and high utility; ACCEPTABLE = minor privacy or utility issues but still usable; BAD = clear leak, high re-identification risk, hallucinated sensitive content, major utility loss, severe over-generalisation or task drift.
- `confidence` $\in$ [0,1]: the judge’s self-reported confidence in its labels. Anchors: 0.3 – limited or ambiguous evidence; 0.6 – moderate evidence; 0.9 – strong evidence.

All numerical analyses in §5.3 (quadrant distributions, leak-rate breakdown, multi-issue rate) operate on these fields directly; verdict labels are used only as a coarse summary.

Bias controls. We adopt two bias controls common in LLM-as-judge work. First, every rewrite is shown to the judge in isolation rather than alongside the output of a competing system, so the judge cannot exhibit *position* or *system-identity* bias across our three methods. Second, the judge is never told which system produced `priv_query`; the JSON payload carries only question, `priv_query`, `ea_items` and `ne_items`, with no system tag. To discourage *verbosity* bias, the rubric explicitly states that preservation can take the form of paraphrases or generalised wording and that issues should be raised only when usefulness is materially reduced.

Output schema. The judge is required to return exactly the following JSON object, with no surrounding prose. The parser used in our analysis pipeline rejects any other shape.

Judge output schema.

```
{
  "privacy_issues": [
    {
      "type": "direct_leak
        | semantic_leak
        | inferential_leak
        | combination_leak",
      "span": "substring from priv_query",
      "source": "sensitive info",
      "reason": "brief explanation"
    }
  ],
  "utility_issues": [
    {
      "type": "lost_constraint
        | weakened_constraint"
```

```
        | task_drift
        | over_generalization
        | added_constraint",
      "span": "substring from priv_query",
      "source": "ea_item or task need",
      "reason": "brief explanation"
    }
  ],
  "new_sensitive": [
    {
      "span": "substring from priv_query",
      "reason": "why not entailed"
    }
  ],
  "reid_risk": 0.0,
  "utility_score": 1.0,
  "verdict": "good | acceptable | bad",
  "confidence": 0.0
}
```

Full judge prompt. The system prompt below is the verbatim instruction we send to GPT-4O-MINI. The user message is a JSON dump of {question, `priv_query`, `ne_items`, `ea_items`} together with the line “Evaluate the following privacy rewrite. Return ONLY the JSON object specified in the system instructions.”

LLM-as-a-Judge system prompt.

You are evaluating whether a privacy rewrite turns a user’s original request into a privacy-preserving query for external task support.

Your goals:

1. Check whether `priv_query` leaks sensitive information from question.
2. Check whether `priv_query` preserves the essential task attributes in `ea_items` from a semantic perspective.
3. Identify utility loss *only* when an essential task attribute is missing, weakened, distorted, over-generalised, or changed in a way that would reduce the usefulness of the external response.
4. Identify newly introduced sensitive information not supported by question.

Use `ea_items` as the primary reference for utility. For each `ea_item`, decide whether it is still present in `priv_query` in a semantically equivalent and task-useful form. Exact wording is not required: paraphrases, abbreviations, generalised forms, split mentions, or implications by a more specific retained detail can count as preserved. Use question only to understand context, resolve ambiguity, and catch clearly indispensable task needs that are missing from `ea_items`. Do not freely invent extra utility requirements beyond `ea_items`.

Use `ne_items` as weak hints for privacy-sensitive information. Do not mechanically match `ne_items`, but pay attention to whether `priv_query` reveals sensitive identifiers, private attributes, or identifying combinations from question, especially when those details are not part of `ea_items`.

Privacy issue types:

- `direct_leak`: `priv_query` directly includes a sensitive identifier or near-verbatim sensitive detail.
- `semantic_leak`: sensitive information preserved through paraphrase, translation, abbreviation, normalisation, or equivalent wording.
- `inferential_leak`: sensitive information made reasonably inferable from common knowledge.
- `combination_leak`: multiple retained clues jointly narrow the user, patient, organisation or situation to a small identifiable set; the combination must create meaningful re-identification risk.

Utility issue types:

- `lost_constraint`: an `ea_item` or indispensable task need is absent.
- `weakened_constraint`: an `ea_item` is present but materially less specific.
- `over_generalization`: `priv_query` is too broad and would elicit only generic background.
- `task_drift`: `priv_query` asks for a different kind of support than the original task requires.
- `added_constraint`: `priv_query` introduces a new task constraint, assumption, diagnosis, preference or framing not supported by question.

Important rules. For multiple-choice questions, answer options need not be preserved unless `ea_items` explicitly require them. Before flagging `lost_constraint` or `weakened_constraint`, ask whether the external source would fail to provide useful task-grounded support because of this omission; if not, do not flag it. Evidence spans must be exact substrings from `priv_query` when possible. Output *only* the JSON object specified above; return no text outside the JSON.